

A NOTE ON ULTRAPRODUCTS

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This note is taken from the first chapter of the book *Ultracategories* by Lurie.

ULTRAFILTER

An ultrafilter on a set S is a boolean algebra homomorphism $\mu: P(S) \rightarrow \{0, 1\}$.

For an ultrafilter μ on S , we can consider the poset $U_\mu := \{A \subseteq S \mid \mu(A) = 1\}$, ordered by inclusion. We can see U_μ as a thin category.

CATEGORICAL ULTRAPRODUCT CONSTRUCTION

Consider a category M^+ with a full subcategory $M \subseteq M^+$.

M has ultraproducts in M^+ if, for every collection $\{M_s \mid s \in S\}$ of objects $M_s \in \text{obj } M$,

- their product $\prod_S M_s$ exists in M^+
- for every ultrafilter μ on S , the diagram

$$\Delta_\mu: U_\mu^{\text{op}} \longrightarrow M^+ \\ A \longmapsto \prod_{s \in A} M_s$$

admits a colimit in M^+ and this colimit belongs to M .

Define the ultraproduct of the objects M_s wrt. the ultrafilter μ as this colimit,

$$\int_S M_s d\mu := \text{colim } \Delta_\mu.$$

ex A category M with small products and small filtered colimits (ones over diagrams where every finite diagram in the domain has a cocone) has ultraproducts in itself.

ex Ultraproduct of sets.

Consider $M = M^+ = \text{Set}$, an ultrafilter μ on a set S and a family of sets $\{M_s \mid s \in S\}$. If all M_s are not empty, then the projections $\pi_A: \prod_S M_s \rightarrow \prod_A M_s$ are surjective and so is the canonical map $q_\mu: \prod_S M_s \rightarrow \int_S M_s d\mu$. Then, we can characterise the ultraproduct as a quotient

$\int_S M_s d\mu \simeq \prod_S M_s / \sim_\mu$
by the equivalence relation
 $(x_s \mid s \in S) \sim_\mu (y_s \mid s \in S) \Leftrightarrow \mu(\{s \in S \mid x_s = y_s\}) = 1.$

ex Extended reals.

Consider the cofinite ultrafilter χ on the set of natural numbers $\mathbb{N}$.

The set of extended reals is the ultraproduct of $\{\mathbb{R} \mid m \in \mathbb{N}\}$ wrt. χ in Set , ${}^*\mathbb{R} := \int_{\mathbb{N}} \mathbb{R} d\chi$.

Overloading notation, the ordered field of extended reals is the ultraproduct of $\{\mathbb{R} \mid m \in \mathbb{N}\}$ wrt. χ in OField .

ULTRAPRODUCT FUNCTOR

Consider a category M with ultraproducts in M^+ and a set S . For a collection $\{f_s: M_s \rightarrow N_s \mid s \in S\}$ of morphisms in M , denote with $\int_S f_s d\mu: \int_S M_s d\mu \rightarrow \int_S N_s d\mu$ the unique morphism given by the universal property of $\int_S M_s d\mu$.

This defines a functor $\int_S (-) d\mu: M^S \rightarrow M$.

By construction, for all A st $\mu(A) = 1$, the morphisms $q_{\mu(A)}^A: \prod_A M_s \rightarrow \int_S M_s d\mu$ form a natural transformation.